Smart Incentive Spirometer- Sprint-3

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Github LInk: https://github.com/Roshan-o/Spirometer

During this sprint, we met with our saroja mam every week after class and worked closely with her to understand the project requirements. We discussed the working principle of the original incentive spirometer and helped us understand the hardware components needed for our project.

We also discussed the overall system design, the sensors required, and the implementation plan. At the same time, we coordinated with the ECE store and Prasanth Sir to arrange the required electronic components. Since some components were not available in the lab, we started by developing the initial version of the website so that software development could continue in parallel.

Also mam suggested that we perform a detailed literature review before starting the hardware implementation. We studied how music is currently integrated into different medical and therapeutic devices, how music helps improve repetitive actions and patient motivation, and the importance of music in healthcare and rehabilitation. This research helped us understand existing approaches, identify useful ideas, and connect them with our Smart Incentive Spirometer project before moving to the implementation phase.

SpiroMeter & It's Working(Original Device):

What is this device?

A 3-ball incentive spirometer is a breathing exercise device used to encourage slow, deep inhalation. It has a mouthpiece connected by tubing to a plastic body containing 3 clear chambers. Each chamber has a lightweight ball.

When a person breathes in through the mouthpiece, air moves through the chambers and lifts the balls. The movement gives the patient immediate visual feedback about the inspiratory airflow being produced.

This type of device is flow-oriented. The balls respond to airflow during inhalation, so they should not be treated as a direct measurement of total lung capacity or as a replacement for a diagnostic spirometry test.



Why is it useful?

After surgery, injury, anesthesia, or a period of limited movement, people may take shallow breaths because of pain, weakness, or reduced activity. Regular deep-breathing practice can help the lungs expand more fully and can support ventilation and secretion clearance.

- Encourages slow, deep breaths.
- Provides a clear visual target during breathing practice.
- Helps maintain lung expansion when activity is limited.
- Can be used as part of breathing care after some surgeries or illnesses, when recommended by a healthcare professional.

Background: how does it help lung expansion?

During a deep breath, the diaphragm contracts and moves downward while the chest expands. This lowers pressure inside the chest, allowing air to enter the lungs.

Shallow breathing gives less expansion to the lungs. A 3-ball incentive spirometer encourages repeated deep inspirations, which helps the patient practice fuller lung expansion. The exercise can be especially useful during recovery when normal deep breathing is difficult.

The device does not permanently increase the physical size of healthy lungs. Its main role is to guide breathing practice and encourage adequate inspiration.

How often is it used?

The schedule depends on the patient's condition and the instructions from a doctor, nurse, or respiratory therapist. A common hospital instruction after surgery is about 10 slow, deep breaths every hour while awake, but the exact schedule can differ.

The target may also change with recovery. Pain, fatigue, age, posture, respiratory condition, and the patient's ability to perform the technique can all affect the result.

Who may use it?

It is commonly used for patients who need help maintaining deep breathing or lung expansion, especially during recovery.

- Patients recovering from chest or abdominal surgery.
- Patients recovering from some illnesses or chest injuries.
- Patients who have been on prolonged bed rest or have reduced activity.
- Patients with reduced ability to take deep breaths, when a healthcare professional recommends it.

How does the 3-ball spirometer work?

The complete process starts when the patient inhales through the mouthpiece.

Step 1: Sit upright and hold the device vertically

The patient sits upright as comfortably as possible and keeps the spirometer in the correct upright position. The mouthpiece is placed firmly between the lips to reduce air leakage.

Step 2: Breathe out normally

The patient first breathes out normally. The exercise then focuses on the next breath, which should be a slow and deep inhalation.

Step 3: Inhale through the mouthpiece

When the patient inhales, pressure inside the chest falls and air is drawn from the room through the mouthpiece and tubing toward the lungs.

Step 4: Air passes through the 3 chambers

The incoming air enters the device and passes through its internal flow path. Each chamber is designed to respond to a particular range of inspiratory airflow.

Step 5: The balls rise

The airflow creates an upward force on the lightweight balls. When the airflow reaches the level needed for a chamber, its ball rises. Common 3-ball models are marked around 600, 900, and 1,200 mL/sec, although the exact values depend on the device.

As the inhalation becomes stronger, additional balls can rise. The balls therefore give a visual indication of the airflow reached during that breath.

Why do the balls rise one by one?

The chambers have different flow requirements. The first chamber responds at a lower airflow, while the next chambers require progressively higher airflow in common 3-ball designs.

So, at a lower inhaled flow, the first ball can rise. With higher flow, the second can rise as well. At still higher flow, the third can rise.

The balls are therefore acting as visual flow indicators. They do not mean that a fixed amount of air has entered the lungs simply because a particular ball is raised.

Inhalation → airflow through chambers → balls rise → visual feedback → deep breath completed → exhalation → balls

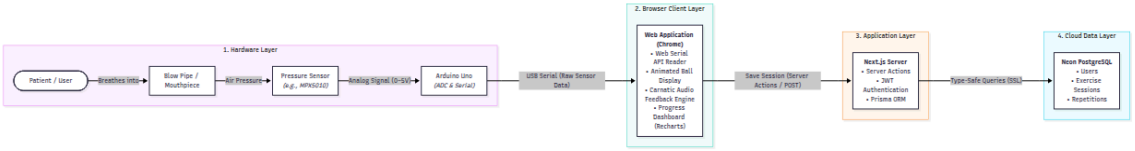
Factors that affect ball movement

- Inspiratory airflow: faster or stronger inhalation can raise the balls further.
- Device design: chamber flow requirements differ between models.
- Patient technique: posture, mouthpiece seal, and breathing pattern affect airflow.
- Respiratory condition: pain, weakness, airway problems, or reduced lung function can limit the achieved flow.
- Device position: the spirometer should be held as directed, usually upright.

References

- • Cleveland Clinic – Incentive Spirometer: Purpose, Goals & How To Use
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- https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm?id=K133873
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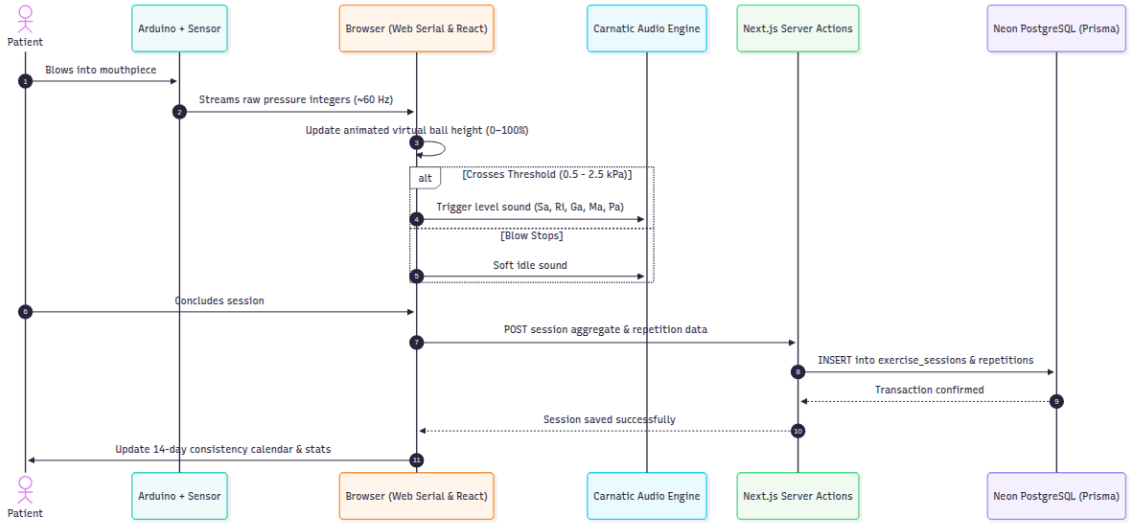
System Architecture:



Figma Link:

https://www.figma.com/design/ZbQEwTaLI2R1L9XhwzSnIZ/Music?t=hc86C48bEpRClTkQ-1

End-to-End Flow:



Identification components:

Initially we have discussed with Ma’am about the Spirometer and Scope of the project, then we finalized the features, scope, and music that we can play at different levels of balls in the spirometer. We have explored various components such as Sensirion SDP31,Sensirion SDP810, BMP388 and came to the conclusion after considering the price of each component,requirements.

Component	Technical Name	Purpose
Microcontroller	ESP32 Development Board	Sensor reading + communication
Sensor	MPX5010DP Differential Pressure Sensor	Measures breathing pressure
Mouthpiece	Medical Spirometer Mouthpiece	User breathing interface
Tube	Silicone Medical Tubing	Air connection
Breadboard	Solderless Breadboard	Circuit prototyping
Wires	Jumper Wires	Electrical connections
Power	5V USB Power Supply/Power Bank	Powers the system

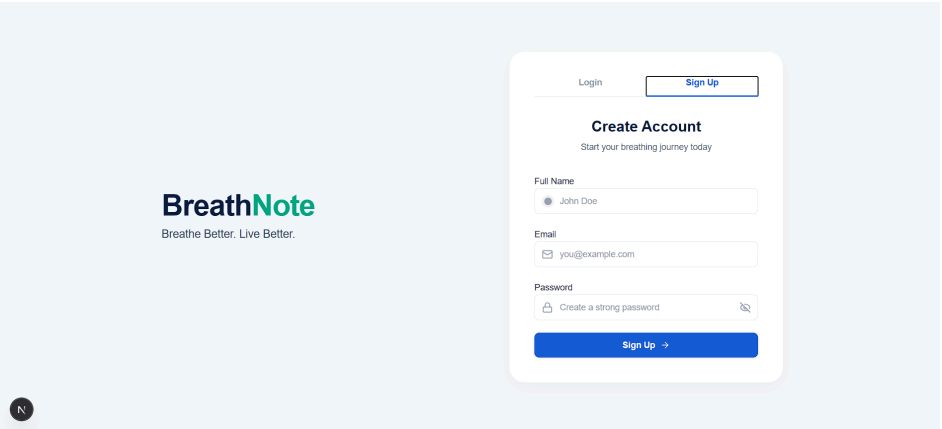
Website Overview

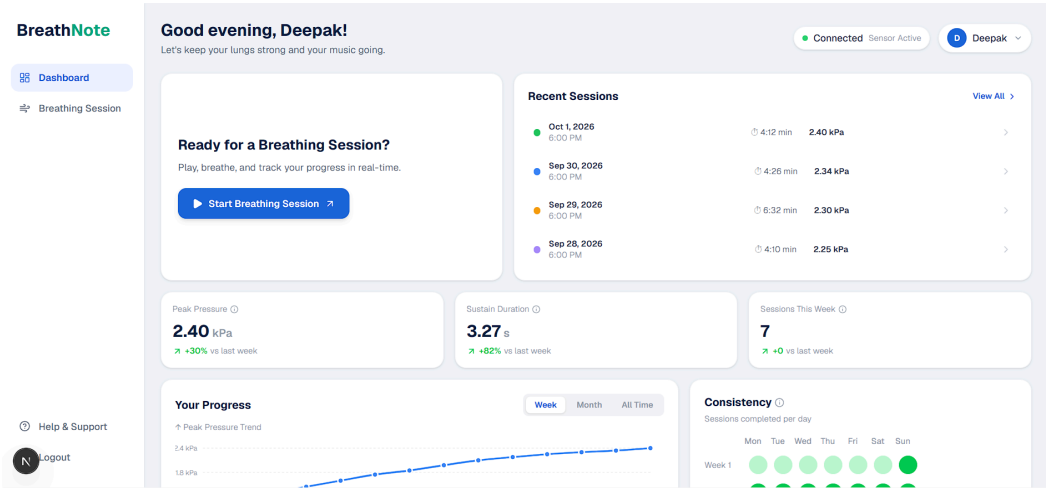
BreathNote is a web-based breathing exercise platform designed to support guided breathing exercises and track user progress over time. The application provides secure user authentication and a personalized dashboard for viewing exercise-session statistics.

The system is implemented using **Next.js** with the **App Router**, **PostgreSQL** for persistent data storage, and **Prisma ORM** for database interaction. Authentication is implemented using password hashing with **bcryptjs** and a seven-day HTTP-only session cookie.

The current implementation consists of two categories of functionality:

1. **Implemented functionality** – authentication and progress visualization.
2. **Designed functionality** – sensor-based breathing-session capture, audio feedback, and session persistence.





Features:

Implemented Features:

User Account and Authentication

The application provides user account management through:

- User signup
- User login
- User logout
- Password hashing using bcrypt
- Session-based authentication
- Seven-day HTTP-only session cookies

During signup, the user's password is never stored directly. Instead, it is converted into a bcrypt hash using ten hashing rounds before being stored in the database.

After successful authentication, the application creates a session cookie with a validity period of seven days. The cookie is configured as HTTP-only so that it cannot be directly accessed by client-side JavaScript.

Progress Dashboard

The dashboard provides a personalized view of the user's breathing-exercise history.

The dashboard currently displays:

- Peak pressure achieved

- Maximum pressure sustain duration
- Number of sessions completed per week
- Pressure trend over time
- Fourteen-day consistency information
- Up to four recent exercise sessions

The dashboard retrieves exercise-session records from PostgreSQL using Prisma.

Per-User Data Isolation

Exercise data is associated with individual users through the user_id field in the exercise_sessions table.

When the dashboard is loaded, the application verifies the session cookie and obtains the authenticated user's ID. Database queries are then restricted to that user ID.

This ensures that a signed-in user can access only the exercise-session records associated with their account.

Application Routes and Server Actions

The application uses the **Next.js App Router** and **Server Actions**. It does not currently expose conventional REST APIs under the `app/api` directory.

Server Actions are framework-managed server-side functions and should therefore be distinguished from conventional REST API endpoints.

Application Routes

Route	Type	Description
/	GET Page	Displays the default application landing page.
/login	GET Page	Displays the login and signup interface.
/dashboard	GET Page	Verifies the session cookie and retrieves the authenticated user's exercise-session data.
/exercise	Planned	Intended to contain the breathing-exercise interface but is not currently implemented.

If the `/` page does not receive a valid session cookie, it currently renders with empty or default dashboard data instead of redirecting the user to the login page.

Database Design

The BreathNote database uses **PostgreSQL** as the relational database management system.

Prisma ORM is used by the application to define the database schema and perform database operations. The database separates user identity information from exercise activity.

The primary relationships are:

User → Exercise Sessions → Repetitions

A single user can have multiple exercise sessions, and each exercise session can contain multiple repetitions.

UUIDs are used as primary keys to uniquely identify database records. Timestamp fields use TIMESTAMPTZ, while pressure measurements use DECIMAL(10,2) to preserve two decimal places of precision.

Prisma model fields use camelCase in the application code and are mapped to snake_case database columns.

Users Table

The users table stores account and authentication information.

Column	Data Type	Constraints / Purpose
id	UUID	Primary key; generated automatically
name	VARCHAR(255)	Stores the user's display name
email	VARCHAR(255)	Required and unique; used for authentication
password_hash	VARCHAR(255)	Required bcrypt password hash
created_at	TIMESTAMPTZ	Stores account creation time
updated_at	TIMESTAMPTZ	Automatically updated when the record changes

The email field is unique to prevent multiple accounts from being registered with the same email address. **Passwords are stored only as bcrypt hashes rather than plaintext passwords.**

Exercise Sessions Table

The exercise_sessions table stores information about individual breathing-exercise sessions.

Column	Data Type	Constraints / Purpose
id	UUID	Primary key; generated automatically
user_id	UUID	Foreign key referencing `users.id`
start_time	TIMESTAMPTZ	Start time of the exercise session
end_time	TIMESTAMPTZ	End time of the exercise session
duration_seconds	INTEGER	Total active duration of the session
max_pressure_achieved	DECIMAL(10,2)	Maximum pressure achieved, measured in kPa
max_level_reached	INTEGER	Highest breathing level achieved
max_level_hold_duration	INTEGER	Duration for which the maximum level was maintained, in milliseconds
number_of_repetitions	INTEGER	Number of repetitions performed; defaults to zero
created_at	TIMESTAMPTZ	Record creation timestamp

The user_id column establishes the relationship between a user and their exercise sessions.

A user can therefore have multiple exercise sessions.

Repetitions Table

The repetitions table stores individual repetition-level information within an exercise session.

Column	Data Type	Constraints / Purpose
id	UUID	Primary key; generated automatically
session_id	UUID	Foreign key referencing `exercise_sessions.id`
timestamp	TIMESTAMPTZ	Time at which the repetition was recorded
max_pressure	DECIMAL(10,2)	Maximum pressure achieved during the repetition, in kPa
level_reached	INTEGER	Highest level achieved during the repetition
hold_duration	INTEGER	Duration for which the level was held, in milliseconds

Each repetition belongs to exactly one exercise session.

Future Work

- Complete the development and integration of the web application, including the virtual ball-level visualization, personalized progress dashboard.
- Develop and assemble a basic hardware prototype consisting of the airflow/pressure sensor, microcontroller, mouthpiece, and connecting tubing.
- Conduct initial testing with a single user to verify the accuracy and consistency of the sensor readings.
- Establish communication between the hardware and web application so that the user's real-time airflow/breathing effort is transmitted to the website.
- Implement and test the virtual ball-level mechanism to ensure that the balls move between predefined levels according to the airflow generated by the user.
- Validate the complete hardware–software integration through basic breathing exercise sessions and refine the pressure/airflow thresholds based on the test results.

Literature Review & Research:

1. Analysis of Music-Based Medical and Therapeutic Devices

Music is increasingly being integrated into medical devices, rehabilitation systems, and digital therapeutics because of its positive effects on both the mind and the body. It helps improve patient motivation, emotional well-being, and overall treatment experience. Research shows that music activates the brain's reward system by releasing dopamine, a neurotransmitter associated with pleasure and motivation, while also reducing stress hormones such as cortisol and adrenaline. In addition, music influences the autonomic nervous system by slowing breathing, reducing heart rate, and lowering blood pressure, which helps patients feel more relaxed during therapy.

In therapeutic and rehabilitation settings, music provides rhythmic cues and real-time feedback that encourage patients to perform repetitive exercises more consistently. These rhythmic patterns improve coordination, support neuroplasticity, and promote relaxation during treatment. Studies have also reported that music therapy can improve memory, communication, and motor performance while reducing muscle tension and increasing pain tolerance. By making therapy more enjoyable and engaging, music helps patients stay motivated and adhere to long-term rehabilitation programs. These combined psychological and physiological benefits have made music an effective component of many modern medical and therapeutic devices.

1. Lenire: Bimodal Neuromodulation for Chronic Tinnitus

Lenire is a non-invasive medical device used to treat **chronic tinnitus**, a condition in which people hear ringing or buzzing sounds without any external source. The device combines specially designed sound played through wireless headphones with mild electrical stimulation delivered to the tongue. These two types of stimulation work together to encourage the brain to reorganize its neural pathways, helping reduce the perception of tinnitus instead of simply masking the unwanted sound.

Sound plays an important role in the therapy. Lenire uses personalized tones and sound patterns that are synchronized with the electrical stimulation to improve treatment effectiveness. The sounds are designed to be comfortable and relaxing while activating specific auditory pathways in the brain. Clinical studies have shown that **80-91% of patients** experienced significant improvement in tinnitus symptoms after treatment, demonstrating that synchronized sound stimulation can be effectively integrated into medical devices to improve therapeutic outcomes.

2. MedRhythms: Rhythmic Auditory Stimulation

MedRhythms is a digital rehabilitation system designed to help people with **stroke** and **Parkinson's disease** improve their walking ability. The system uses small sensors attached to the patient's shoes to measure walking patterns such as walking speed, step length, and balance. This information is sent to a mobile application, which adjusts the music in real time based on the patient's movement. As the patient walks, the music changes to guide and encourage better walking performance.

The system mainly uses rhythm instead of melody. The steady beat acts like an external metronome, helping patients match their footsteps with the music and improve coordination, balance, and walking speed. Clinical studies showed that stroke patients improved their walking speed by about **22%**, while Parkinson's disease patients increased their daily physical activity by around **28%**. This shows that real-time music feedback can improve rehabilitation by motivating patients and encouraging repetitive movements. The same idea is used in our proposed **Smart Incentive Spirometer**, where Carnatic music changes according to the patient's breathing performance to encourage correct and consistent breathing exercises.

3. Pacifier Activated Lullaby (PAL): Operant Conditioning in Neonatal Care

The **Pacifier Activated Lullaby (PAL)** system is a medical device used in Neonatal Intensive Care Units (NICUs) to help premature infants learn proper feeding skills. Premature babies often cannot coordinate sucking, swallowing, and breathing, making oral feeding difficult. The device consists of a pacifier with a built-in pressure sensor that measures the infant's sucking strength. When the baby sucks with enough pressure, the device immediately plays a soothing lullaby. If the sucking stops or becomes weak, the music also stops. This immediate musical reward encourages the infant to continue practicing the correct sucking pattern and gradually improves feeding ability.

The PAL system shows how music can be used as real-time positive feedback instead of simple entertainment. The lullabies are soft, predictable, and played at a safe volume to keep infants calm while motivating them to perform the desired action repeatedly. Clinical studies have shown that this approach helps premature infants develop feeding skills faster and reduces the time they spend in the hospital. This concept is similar to our proposed **Smart Incentive Spirometer**, where Carnatic music is played only when the patient performs the breathing exercise correctly, encouraging proper technique and improving motivation during respiratory rehabilitation.

4. Sensate: Infrasonic Resonance and Autonomic Regulation

Sensate is a wearable device designed to reduce **stress, anxiety, and sleep problems** by helping the body relax. The device is placed on the chest and produces gentle low-frequency vibrations while the user listens to specially designed calming audio through headphones. The vibrations and sound work together to stimulate the vagus nerve, which helps shift the body from a stressed state to a relaxed state. The therapy is simple to use and usually lasts between 10 and 30 minutes.

The music used in Sensate is slow, calming, and synchronized with the vibrations produced by the device. Instead of fast rhythms or loud sounds, it uses deep tones, soft harmonies, and nature-inspired soundscapes to promote relaxation. Studies have shown that regular use of Sensate can reduce stress, improve sleep quality, and help users relax more quickly. This demonstrates how synchronized music and physical feedback can be combined in medical devices to improve well-being.

5. Emerging Acoustic Modalities: Raga Therapy and Microtonal Interventions

Raga therapy is an emerging approach that uses **Indian classical music**, including Carnatic and Hindustani ragas, to improve physical and mental health. Unlike many existing medical devices that use Western music or simple rhythmic sounds, raga therapy uses specific melodic patterns and microtones (*shrutis*) to influence emotions, relaxation, and the autonomic nervous system. Research suggests that different ragas can help reduce stress, anxiety, depression, insomnia, and high blood pressure by promoting relaxation and improving emotional well-being.

Although raga therapy has not yet been widely integrated into commercial medical devices, studies show that it has strong potential for future healthcare applications. Different ragas are associated with different therapeutic effects and may improve heart rate variability, vagal activity, and overall mental relaxation.

From the literature review, **we studied medical and therapeutic devices that integrate music in different ways, including tinnitus treatment, gait rehabilitation, neonatal care, stress management, and emerging raga-based therapies.** Although these devices target different medical conditions, they all demonstrate that music is much more than a source of

entertainment. Music can act as a therapeutic tool by increasing patient motivation, encouraging repetitive actions, providing real-time feedback, improving engagement, promoting relaxation, reducing stress and anxiety, enhancing neuroplasticity, supporting motor coordination, improving patient adherence to therapy, and making long-term rehabilitation more enjoyable. Many of these systems also synchronize music with a patient's physiological activity, allowing music to respond to the user's performance and provide immediate positive reinforcement.

These findings strongly support our proposed **Smart Incentive Spirometer**, where Carnatic music is integrated with breathing exercises to provide real-time feedback, motivate patients to perform correct inhalation, improve adherence to respiratory therapy, and create a more engaging and effective rehabilitation experience.

2.Effect of Music on Repetitive Actions and Its Application in Respiratory Rehabilitation

The objective of this literature review was to understand how music influences repetitive human activities, particularly its effects on attention, motivation, boredom, and task engagement.
Since patients using an incentive spirometer must repeatedly perform the same breathing exercise several times every waking hour, maintaining long-term motivation becomes a major challenge.
By reviewing studies on repetitive actions, music therapy, neuroscience, and Carnatic music, we aim to identify how music can improve patient engagement and apply these findings to the design of our **Smart Incentive Spirometer**, where adaptive Carnatic music is used to encourage correct breathing, reduce monotony, and improve therapy adherence.

The Problem with Repetitive Actions and Monotony

The human brain is designed to learn new things, solve problems, and respond to changes in the environment. It naturally looks for variety and new experiences to stay active.

When a person performs the **same task repeatedly** every day without any change or interest, the brain receives less stimulation. First, repetition helps people learn a skill and perform it faster. However, doing the same activity for a long time without engagement can have negative effects.

Over time, repetitive and monotonous work can reduce attention, lower motivation, and make people feel mentally tired. It can also **decrease productivity, increase boredom, and affect emotional well-being.**

Keeping people engaged through variety, meaningful activities, or enjoyable experiences is important because it helps maintain focus, motivation, and overall mental health.

The Psychological Effects of Repetitive Work

Boredom is often seen as a small problem, but it is actually an important signal from the brain. It tells us that our activities are no longer interesting or challenging. When we keep doing the same task every day without any change, the work becomes predictable and less enjoyable.

If this continues for a long time, temporary boredom can become **chronic boredom**. This can lead to stress, frustration, anxiety, low motivation, and even depression. People may start feeling disconnected from their work and lose interest in what they do.Psychologists call this condition **bore-out** and this can seriously affect a person's mental and physical health.

The Neurological Effects of Repetitive Work

The brain releases a chemical called **dopamine** when we do something new, interesting, or challenging. Dopamine helps us stay motivated, focused, and interested in our work.When we keep repeating the same task for a long time, the brain gets used to it. As a result, dopamine levels decrease, making the person feel less motivated, mentally tired, and uninterested.

The brain also follows the rule of **"use it or lose it."** If we do not regularly use our thinking and problem-solving skills, the brain starts relying on automatic habits instead of active thinking. This reduces mental alertness and makes it harder to stay focused.

People who already have attention difficulties, such as **ADHD**, may find repetitive tasks even more challenging because they need more mental stimulation to stay engaged.

Long-term boredom can also affect physical health. It may increase stress levels, encourage unhealthy habits, reduce attention, and contribute to problems such as headaches, stomach discomfort, and even heart-related diseases. Throughout history, people have recognized this feeling of long-lasting boredom and mental exhaustion as a serious condition that affects both the mind and the body.

Incentive Spirometry: A Repetitive Medical Exercise

The negative effects of repetitive actions are not limited to everyday work. They are also seen in medical treatments, especially during lung rehabilitation. One common example is the **incentive spirometer**, a device that helps patients improve their breathing after surgery or during recovery from lung diseases.

An incentive spirometer is widely used after **chest, heart, or abdominal surgery**. It is also recommended for people who have been on bed rest for a long time or have lung conditions such as **COPD**.

The device helps prevent the collapse of tiny air sacs in the lungs (alveoli) and improves overall lung function.

How an Incentive Spirometer Works

The incentive spirometer works on the principle of **Sustained Maximal Inspiration (SMI)**, which means taking a **slow, deep breath and holding it for a few seconds**.

The patient places the mouthpiece in the mouth and inhales slowly. As they breathe in, the device provides visual feedback by lifting a piston or balls, showing how much air has been inhaled.

Taking a deep breath expands the lungs and opens the tiny air sacs (alveoli). This improves oxygen flow and helps prevent lung complications such as atelectasis (collapsed air sacs).

After inhaling deeply, the patient holds their breath for **3-5 seconds** before breathing out normally. This gives the lungs enough time to fully expand.

To get the best results, doctors usually recommend repeating this exercise **5-10 times every waking hour**. Although the exercise is simple, repeating it many times each day can become boring, causing some patients to lose motivation and skip their breathing exercises. This reduction in patient engagement can slow down the recovery process.



The detailed analysis for working of the Incentive spirometer is already covered.

The Problem with Patient Compliance

Although the incentive spirometer is very important for recovery, many patients find it difficult to use regularly because the exercise is repetitive and boring. They have to perform the same breathing action many times every day without any change or interaction.

Since the activity does not provide enough mental stimulation, patients quickly lose interest. Instead of focusing on the exercise, they often think about their pain, discomfort, or stress after surgery. As a result, they may skip or stop using the device.

Studies have shown that many patients do not use the incentive spirometer correctly, and a large number stop using it completely during recovery. Poor compliance reduces the effectiveness of the treatment and increases the risk of lung complications after surgery.

This shows that medical devices should not only be effective but also engaging. Adding interactive and motivating features can encourage patients to complete their breathing exercises more consistently.

Music as a Therapeutic Solution

Researchers have found that **music** can make repetitive activities more enjoyable and engaging. Unlike repetitive tasks, music actively stimulates different parts of the brain.

Listening to music activates areas responsible for **hearing, attention, memory, emotions, and movement** at the same time. This keeps the brain interested and reduces the feeling of boredom.

Because music improves mood and motivation, it can encourage patients to continue their breathing exercises for a longer time. By making the therapy more enjoyable, music has the potential to increase patient compliance and improve recovery outcomes.

How Music Affects the Brain and Body

Music does more than entertain us-it can change how our brain and body function. Researchers have found that listening to music can influence **brain waves**, which are the electrical signals produced by the brain.When people are stressed, anxious, or bored, the brain is often in a **Beta wave** state. This state is linked to high alertness, worry, and difficulty concentrating.

Calm and relaxing music can shift the brain into **Alpha** and **Theta** wave states. These states help people feel relaxed, focused, and emotionally calm. They also reduce stress, anxiety, and excessive thinking.

Music also affects the **autonomic nervous system**, which controls automatic body functions such as heart rate and breathing. Relaxing music stimulates the **vagus nerve**, which activates the body's natural relaxation response.

As a result, music can lower stress hormone (cortisol) levels, reduce heart rate, and improve **Heart Rate Variability (HRV)**. A higher HRV is a sign of better emotional health, lower stress, and a healthier nervous system.

When breathing is synchronized with slow, steady music (**~ 6 breaths per minute**), the body relaxes even more. This improves breathing efficiency, supports heart health, and creates a

calming effect, making it especially useful during breathing exercises such as incentive spirometry.

Research Findings on Music Therapy

Many recent studies have shown that **music therapy** can improve both mental and physical health during medical treatment. Researchers have found that listening to music during recovery helps patients feel calmer, reduces stress and anxiety, and improves their overall hospital experience.

Studies also show that music therapy can shorten hospital stays and reduce the need for certain pain-relief and anesthesia medications during medical procedures.

With advances in technology, music therapy is now combined with **Virtual Reality (VR)** to create relaxing and immersive environments. Patients can listen to calming music while experiencing peaceful virtual scenes, helping them forget the stressful hospital surroundings.

One study found that patients who used **VR with Indian classical (Raga) music** for several days experienced lower stress, anxiety, and depression. They also showed signs of better relaxation and healthier nervous system activity.

Effects of VR-Based Music Therapy

Parameter	Result After Music Therapy	Benefit
Stress Level	Decreased	Patients felt more relaxed and less anxious.
Heart Rate	Became more stable	Improved relaxation and cardiovascular health.
Breathing Rate	Decreased naturally	Breathing became slower and more comfortable.
Nervous System Activity	Improved	Increased relaxation through greater parasympathetic activity.

Carnatic Music and Raga Therapy (Raga Chikitsa)

Raga Therapy (Raga Chikitsa) is a traditional Indian healing method that uses **Carnatic** and **Hindustani classical music** to improve mental and physical health. Different **ragas**

create different emotional and physiological effects, helping people feel calm, relaxed, energetic, or focused.

A **raga** is more than just a musical scale. It is a combination of musical notes (**swaras**) arranged in a specific order with unique ascending (**Arohana**) and descending (**Avarohana**) patterns. Each raga also follows a particular rhythm (**Tala**) and expresses a specific mood or emotion.

Researchers have found that certain ragas can help reduce **stress, anxiety, high blood pressure, asthma, and chronic pain**. Raga Therapy also considers the **time of the day**. For example, morning ragas are believed to improve alertness and energy, while evening ragas help people relax and sleep better.

Gamakas: The Unique Feature of Carnatic Music

A special feature of Carnatic music is the **Gamaka**, which means adding smooth movements or **ornamentation** to musical notes. Instead of singing a note in a fixed way, the singer gently bends, oscillates, or slides between notes.

There are different types of gamakas, such as **Kampita**, **Nokku**, **Sphurita**, and **Jaaru**. These techniques make the music expressive and pleasant to listen to.

Because the sound is always changing, the brain stays active and interested. Unlike repetitive sounds, gamakas prevent the brain from getting bored. This helps improve attention, concentration, and mental engagement, making Carnatic music useful during repetitive activities such as breathing exercises.

Respiratory Mechanics: Music Helps Control Breathing

Instead of asking patients to sing, they can simply **listen to carefully selected Carnatic music** while using the incentive spirometer.

The slow rhythm (**Tala**) and long, flowing musical phrases naturally encourage **slow and deep breathing**. As patients listen to the music, they tend to inhale more slowly, hold their breath for a few seconds, and exhale in a controlled manner.

This breathing pattern matches the correct technique required for using an **incentive spirometer**. The music acts like a natural breathing guide, making the exercise easier and more enjoyable.

Using Carnatic music during spirometry can **reduce boredom, improve patient motivation, encourage correct breathing, and increase compliance with the therapy**, leading to better recovery and improved lung function.

A 4-Level Carnatic Music for Incentive Spirometry

To make incentive spirometry more interesting and enjoyable, a **web-based platform** can be developed. Patients simply open the website and listen to the recommended Carnatic music while performing their breathing exercises.

The incentive spirometer measures the amount of air inhaled, usually from **500 mL to 2500+ mL**. Based on the patient's lung capacity, the website provides different Carnatic music tracks. As the patient's breathing improves, they can move to the next level.

The protocol mainly uses **Raga Mayamalavagowla**, which is known for creating a calm and focused state of mind. As patients progress, more advanced music is introduced to keep them engaged and motivated throughout their recovery.

Note: The main music to be played is in discussion in saroja mam.The below mentioned raga's are found during our research from the web.

Level 1: Relaxation and Basic Breathing (~ 500-1000 mL)

When the patient begins the breathing exercise, the website continuously receives the spirometer reading. If the inhaled volume is between **500 and 1000 mL**, the system automatically plays the **Level 1** audio.

The Level 1 audio consists of a **continuous Tanpura Sruti** with a soft flute melody in **Raga Mayamalavagowla**.

The continuous Sruti provides a stable and relaxing sound that reduces anxiety and encourages slow, deep breathing. Since patients at this stage have limited lung capacity, simple and calming audio is preferred over complex musical patterns.

Level 2:(1000-1500 mL)

When the spirometer detects an inhalation between **1000 and 1500 mL**, the website automatically switches to the **Level 2** audio.

This level plays **Sarali Varisai** in **Raga Mayamalavagowla** with a slow and steady rhythm. The patient continues breathing in synchrony with the music, using the rhythm to maintain smooth and controlled breathing.Sarali Varisai follows a simple ascending and descending sequence of notes. Its predictable rhythm helps patients maintain consistent breathing while gradually increasing lung capacity.

Level 3:

When the patient consistently reaches **1500-2000 mL**, the website automatically changes to the **Level 3** audio.

The audio consists of **Janta Varisai** in **Raga Kalyani**. The repeated note patterns make the listening experience more engaging and encourage the patient to continue the breathing exercise without losing interest.

Because Janta Varisai requires greater attention because of its repeated note structure. Combined with the calm and uplifting nature of Raga Kalyani, it helps reduce boredom during longer breathing sessions.

Level 4: Maximum Lung Expansion

When the spirometer reading exceeds **2000 mL**, the website automatically plays the **Level 4** audio.

The patient listens to a slow **Alapana** in **Raga Kalyani** while performing deep breathing exercises. The long musical phrases naturally support slow inhalation, breath holding, and controlled exhalation.

Alapana contains continuous melodic variations and long flowing phrases. This prevents monotony, keeps the brain engaged, and makes repeated breathing exercises more enjoyable for patients who have reached the highest level of therapy.

3. Integration of Music into Medical and Therapeutic Devices

Neurorehabilitation and Motor Recovery

Many rehabilitation devices use music to help patients recover after a stroke or brain injury.

One example is the **MusicGlove**. It is a glove with sensors on the fingertips. Patients play a game similar to Guitar Hero by moving their fingers in time with music. This makes therapy more interesting and encourages patients to perform hundreds of hand movements. Research shows that people using MusicGlove improved their hand movement more than people using normal therapy. Music keeps patients motivated, so they practice more often, which helps the brain recover faster.

Music is also used to improve walking and balance. Patients walk according to the beat of music or a metronome. This improves walking speed, balance, and coordination, especially for people with Parkinson's disease or stroke.Some rehabilitation systems use cameras, such as Microsoft Kinect or Nintendo Wii, to turn exercises into games with music. Patients enjoy these games, so they continue therapy for longer periods.

Physiotherapy and Exercise Therapy

Music is also used in physiotherapy through exercise games called **exergames**.Devices like **FitMi** use motion sensors to detect body movements while music plays during exercises. The music and game-like environment make therapy more enjoyable.Research shows that when exercise feels like a game, patients exercise more regularly and put in more effort. This improves recovery and physical fitness.

Music can also improve heart and lung health.Studies show that listening to calm music every day reduces blood pressure, anxiety, stress, and depression. Different types of classical music, including Indian classical ragas, have shown positive effects on heart health.

Some breathing devices, such as electronic spirometers, include music or sound effects during breathing exercises. These sounds encourage patients to breathe correctly and complete their exercises.Music also relaxes patients, making breathing exercises easier and improving treatment results.

Pain, Anxiety, and Stress Management

Music is widely used to reduce pain and stress.During painful medical procedures, soft music helps distract patients from pain and makes them feel calmer. Music can reduce blood pressure, heart rate, and anxiety.Studies in India found that listening to Carnatic ragas every day reduced anxiety, improved sleep, and lowered stress levels.

Modern wearable devices also use music with biofeedback. For example, EEG headbands measure brain activity and play relaxing sounds when the user becomes calm. These devices help people reduce stress and improve sleep.

Cognitive Support and Mental Health

Music also improves brain function.Listening to enjoyable music can improve attention, memory, mood, and concentration. Music therapy combined with normal treatment often reduces depression and anxiety better than treatment alone.

Interactive music activities, such as playing instruments or rhythm games, stimulate different parts of the brain. These techniques are being used to help people with dementia, ADHD, and other neurological conditions.Virtual reality systems are also beginning to include music for cognitive rehabilitation.

Technologies and Sensors Used

Modern music-based medical devices use many different sensors.

Some common sensors include:

- Motion sensors to detect body movement.

- Pressure and touch sensors to detect hand or finger movements.
- EEG sensors to measure brain activity.
- Heart rate sensors.
- EMG sensors to measure muscle activity.
- Respiratory flow sensors to measure breathing.

Benefits of Music-Based Devices

One of the biggest advantages of adding music is improved motivation.Without music, therapy can become repetitive and boring. Music makes therapy enjoyable, encouraging patients to continue practicing every day.Research shows that music increases patient satisfaction, improves exercise consistency, and helps people tolerate pain better.Music also improves mood by increasing dopamine and reducing stress hormones like cortisol. These changes may help the brain learn faster and improve rehabilitation outcomes. **(Detailed review already covered by teammate)**

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